Design and Implementation of an Intelligent Virtual Agent for High Performance Computing Support Ticketing System

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Abstract. High-Performance Computing (HPC) environments, such as the Kabré supercomputer at the National High Technology Center (CeNAT) in Costa Rica, face significant challenges in providing efficient and scalable user support due to the heterogeneity and growing size of their user communities. Traditional ticket-based systems introduce delays, repetitive workload for support staff, and limited interactivity, negatively impacting both user experience and system efficiency.
This work proposes the design, implementation, and evaluation of an Intelligent Virtual Agent (IVA) based on large language models (LLMs), integrated into the Kabré infrastructure to provide real-time technical assistance. The agent is designed to answer common support questions related to job execution, software configuration, and resource management, while routing complex cases that require human intervention to support specialists.
The system leverages natural language processing (NLP) techniques and semantic embeddings derived from the historical ticket database of the Kabré support system. This enables the identification of user requests and the determination of whether they can be resolved automatically by the virtual agent or should be escalated to human support staff.
User interaction is performed exclusively through text input, while system responses are delivered in both textual form and through text-to-speech (TTS) synthesis, providing a multimodal output experience without requiring voice input from the user.
Finally, the system was evaluated through a within-subject experimental study using representative HPC support tasks, comparing the performance of the IVA against users' prior experience with the traditional ticketing system. The evaluation was conducted in terms of efficiency, usability, and cognitive load using objective performance metrics and standardized questionnaires.

Keywords: High-Performance Computing (HPC) · Intelligent Virtual Agent · Large Language Models · Kabré supercomputer · technical support systems · human-computer interaction · conversational agents · user experience.

1 Problem Definition

High-Performance Computing (HPC) environments typically host highly heterogeneous user communities composed of students, researchers, and professionals from diverse disciplines. This diversity naturally leads to a wide range of technical expertise, resulting in queries of varying complexity—from basic questions about system usage and command-line operations to advanced issues involving software installation, resource optimization, and memory management.

In Costa Rica, the Kabré supercomputer, operated by the National High Technology Center (CeNAT), represents a critical infrastructure for scientific research. The system currently supports approximately 300 active users, generating over 100,000 computing hours annually. This community is primarily composed of students, faculty members, and researchers from public universities across the country, spanning multiple domains of knowledge. Despite its importance, user support poses a significant operational challenge, as it is managed by a small team consisting of a single infrastructure engineer and one technical assistant.

At present, user support is handled through a ticket-based system, where users submit queries that may include problem descriptions, code snippets, and supporting files. While functional, this approach presents several limitations. First, it introduces friction in the user experience, as users must navigate the platform, authenticate, and formally structure their requests before receiving assistance. Second, resolving issues often requires multiple back-and-forth interactions to clarify context, which increases response and resolution times.

Furthermore, the current volume of technical support requests can be objectively quantified through the analysis of the historical ticket management records. Data from 2024, 2025, and the first half of 2026 reveal a sustained level of support activity, with fluctuations associated with periods of increased utilization of the HPC infrastructure. In several months, the number of tickets created did not match the number of tickets resolved within the same period, resulting in a temporary accumulation of pending requests (backlog). This behavior is particularly evident during periods of high computational demand, when increased usage of the supercomputer leads to a corresponding rise in support requests related to system access, job execution, software configuration, and resource optimization. Although most tickets are eventually resolved in subsequent periods, these fluctuations reflect a variable operational workload for the support team and highlight the need for mechanisms capable of absorbing peaks in demand without negatively impacting response times.

Although a knowledge base exists within the current system, it is outdated, limited in scope, and lacks interactivity. Users cannot perform targeted queries or obtain immediate answers, forcing them to rely entirely on the ticketing system. As a result, response times can exceed 24 hours, which is particularly problematic in scenarios requiring rapid troubleshooting or near real-time assistance.

Given these challenges, there is a clear need to leverage modern technological solutions to improve user support in HPC environments. In particular, the development of an intelligent virtual agent emerges as a promising alternative for automating the handling of user inquiries, especially those that are repetitive in nature. Such a system could provide real-time assistance, reduce the operational workload of the infrastructure staff, and enhance the overall user experience. However, during its initial stages, the agent should not operate fully autonomously; instead, it should work in collaboration with the infrastructure team, determining which queries can be effectively resolved automatically and which should be escalated to the support staff.

2 Literature Review

Early research in High-Performance Computing (HPC) began to challenge the notion that computational performance alone defines system effectiveness, instead emphasizing the importance of interaction, usability, and human-centered design. One of the earliest contributions in this direction is the work of Faigle, Fox, Furmanski, Niemiec, and Simoni (1993), who established that virtual reality (VR) and HPC are inherently complementary technologies. They argued that large-scale "Grand Challenge" simulations require dynamic VR interfaces capable of visualizing complex computational processes in real time. A key technical insight from their work is the need to decouple the update rates of HPC computational engines from VR display systems to prevent instability and user discomfort, such as simulation sickness [10]. Despite these advances, the authors identified critical limitations, including the lack of standardized connectivity models and the absence of hardware-independent interface protocols. These challenges remain relevant today, highlighting a persistent gap between computational performance and interactive usability that modern HPC systems continue to address [10].

Building on the importance of interaction, Cai and He (2004) introduced an early approach to enhancing human–machine communication through avatar-based interfaces. At the time, most interaction systems relied on rigid rule-based mechanisms and static learning models that were unable to adapt to different users or contexts. To overcome these limitations, the authors proposed the Interactive Agent Based on Awareness (ABI Agent), a system designed to simulate human-like behavior through a structured process of intention generation, goal selection, and action planning. Their results demonstrated that embedding behavioral characteristics into agents can significantly improve user efficiency by automating repetitive tasks and enabling more intuitive interaction.

However, Cai and He (2004) also acknowledged fundamental challenges in achieving truly adaptive and human-like systems. As discussed in [5], integrating personality and human knowledge into intelligent agents remains difficult due to the complexity and variability of psychological and contextual factors across users. Additionally, early systems lacked the ability to learn autonomously from interaction, limiting their capacity to adapt dynamically to user needs. These limitations underscore the broader challenge of moving from static, rule-based systems toward more flexible, learning-driven interaction models.

A more explicit focus on the human dimension in HPC emerges in the work of Monroy, Becerra, Bellas, and Duro (2006, 2008), who argue that traditional optimization strategies are often misaligned with user needs. Rather than prioritizing metrics such as job turnaround time or queue waiting time, they emphasize that the ultimate objective of HPC systems should be to maximize user satisfaction while minimizing inefficiencies. Their studies reveal that users frequently overestimate required computational resources—particularly CPU time—as a precaution against job failure, leading to suboptimal scheduling and reduced system efficiency.

To address this issue, the authors developed the EVOPROC interface, which combines neural networks and evolutionary algorithms to predict actual resource consumption more accurately than manual user submissions [17, 16]. This approach enables HPC systems to interpret imprecise user inputs and translate them into optimized execution plans, effectively bridging the gap between human intent and system-level performance. Furthermore, they identify behavioral indicators such as frequent system queries, including `qstat`, as signals of user uncertainty and impatience, highlighting the importance of incorporating behavioral data into system design.

Subsequent research, including [18], reinforces the importance of user behavior modeling as a critical yet underdeveloped component of HPC scheduling systems. In particular, these studies point out the difficulty of formalizing subjective dimensions such as satisfaction, trust, and perceived responsiveness, which vary significantly across users and contexts. As a result, a persistent gap remains between technical performance metrics and human-centered evaluation.

Moreover, both Monroy et al. and later studies highlight a structural limitation in traditional HPC schedulers: although they provide extensive configuration parameters, there is often no clear mapping between these parameters and the quality of service perceived by users. This disconnect is further exacerbated by users' limited understanding of their own computational requirements. Consequently, there is a growing need for intelligent systems capable of translating uncertain or conservative user inputs into efficient and reliable execution strategies.

Overall, these foundational works mark a critical transition in HPC research—from systems focused purely on computational optimization toward environments that increasingly account for human interaction, perception, and adaptability. While early approaches laid the groundwork for integrating usability into HPC, they also revealed significant limitations that continue to motivate the development of more intelligent, user-aware, and autonomous systems.

To bridge the early human-centered vision of HPC with more mature intelligent interaction systems, recent research begins to formalize how user experience, personalization, and adaptive assistance can be operationalized at scale. While foundational works established the importance of visualization, user satisfaction, and basic agent behavior, the next stage of research shifts toward embedding intelligence directly into support systems, enabling them to interpret user needs, learn from interaction data, and provide context-aware assistance. This transition marks the emergence of a second layer in HPC evolution, where interaction is no longer passive but actively mediated by intelligent, adaptive systems.

Within this context, Biggs and Dalton (2024) highlight a persistent knowledge gap among researchers, emphasizing that expertise in one HPC system does not easily transfer to another. To address this, they demonstrate that integrating AI assistants based on *RAG* enables highly precise, site-specific technical support grounded in continuously updated internal documentation, thus avoiding the costly need for frequent model retraining [2]. However, they also identify a critical limitation: historical support data—such as tickets—often contain noise, including irrelevant conversations or sensitive information, which complicates their transformation into reliable knowledge sources.

This challenge of creating more natural and adaptive interaction systems echoes earlier concerns raised by Cai and He (2004), who argue that traditional avatar-based interfaces lack personality and flexibility. Their proposed ABI Agent represents an important step toward more human-like interaction, introducing a structured process of intention generation, goal selection, and action planning. By embedding behavioral tendencies into agents, their approach demonstrates measurable improvements in user efficiency, particularly in handling repetitive or cognitively demanding tasks. Nevertheless, they acknowledge that fully modeling human personality and contextual understanding remains a complex and unresolved problem, especially given the variability of psychological and sensory factors across users [5].

Expanding on the limitations of interaction design, Diederich et al. (2020) provide a broader perspective on conversational agents, arguing that many systems fail not because of technical shortcomings, but due to their neglect of user-specific characteristics such as culture, personality, and expectations. They advocate for longitudinal, real-world studies to better understand how social and behavioral cues influence long-term user engagement and system sustainability [7]. This reinforces the idea that effective HPC support systems must go beyond functional correctness and incorporate deeper human-centered design principles.

From an operational standpoint, Sanjeewa (2020) situates these challenges within the broader landscape of technical support systems, noting that the growing complexity of enterprise environments places increasing pressure on support infrastructures. The study emphasizes that AI-driven support tools are most effective when deployed in a human-in-the-loop configuration, where chatbots handle repetitive queries while experts focus on complex issues. However, important gaps remain, including the lack of specialized ticketing systems and the need for continuous learning mechanisms driven by real user feedback, as well as enhanced security features such as multi-factor authentication.

Building on these ideas within the HPC domain, Shifare, Shukla, and Kothari (2025) demonstrate how intelligent systems can significantly improve both access management and user support. Their AI-based Laboratory Manager (ALM) reduces resource access approval times from days to minutes, while a fine-tuned RoBERTa-based chatbot achieves a 98.1% accuracy rate in resolving user queries. Despite these advances, the study highlights persistent issues such as ambiguity in user requests and inconsistencies in documentation, while also pointing toward future directions in predictive resource management and more advanced scheduling optimization [24].

Finally, Wulf and Meierhofer (2024) extend this discussion by analyzing the role of large language models in technical customer service environments. They show that while LLMs are highly effective for low-level cognitive tasks such as summarization and translation, more complex problem-solving requires enhanced architectures like *RAG* or fine-tuning to reduce hallucinations. Their work also underscores the importance of structured data ecosystems to enable context-aware AI systems. However, they identify a critical gap in the lack of large-scale, real-world validation studies, as well as challenges related to user acceptance, usability, and integration with legacy systems [25].

Together, these studies define Cluster 2 as a transition toward intelligent, user-centered support systems in HPC, where the focus shifts from static interfaces to adaptive, learning-driven interaction layers capable of bridging the gap between complex computational infrastructures and diverse user communities.

As intelligent interaction layers mature and begin to effectively mediate between users and complex HPC infrastructures, the next logical step is to optimize what happens behind the interface: the allocation, scheduling, and orchestration of computational resources. This shift marks a transition from user-centered assistance toward system-level intelligence, where AI is not only supporting users but actively managing workloads, predicting behavior, and optimizing performance in real time. In this sense, Cluster 3 establishes the operational backbone upon which higher-level automation—such as fully autonomous support systems—can be built.

Within this framework, Fan et al. (2020) demonstrate that traditional scheduling heuristics like FCFS are fundamentally inadequate for dynamic HPC environments, as they fail to adapt to fluctuating workloads and often require continuous manual tuning. Their proposed DRAS system leverages hierarchical deep reinforcement learning to autonomously manage resource reservation and backfilling, achieving performance improvements of up to 45%. However, the authors highlight a critical limitation: models originally designed for cloud computing do not translate well to HPC due to factors such as job rigidity and starvation, revealing an important gap in domain-specific AI design [11].

Complementing this work, Narantuya et al. (2022) expand the optimization perspective by incorporating energy efficiency into the scheduling problem. Their decentralized multi-agent deep reinforcement learning ($mDRL$) approach significantly improves both performance and sustainability, reducing task completion times by 20% and inefficient energy consumption by 40% [19]. Despite these gains, the study exposes structural limitations in current HPC ecosystems, particularly regarding accessibility for smaller institutions and the inability of centralized architectures to meet low-latency demands. This reinforces the need for hybrid models that integrate Edge Computing with traditional HPC infrastructures.

From a behavioral standpoint, Zaborovsky et al. (2023) reveal that inefficiencies in HPC systems are not only technical but also human-driven. Their analysis shows that users overestimate job execution times in more than 89% of cases, leading to poor resource utilization and congested queues. To address this, they propose a multi-agent scheduling system enhanced with Random Survival Forests (RSF), which predicts probabilistic execution times rather than fixed estimates. While this improves accuracy, the authors emphasize the necessity of Explainable AI (XAI) to ensure that users understand system decisions, and they advocate for the integration of Natural Language Processing (NLP) interfaces to facilitate more intuitive interaction with scheduling systems [27].

In parallel, Jain, Gupta, and Neha (2024) tackle inefficiencies from the perspective of support infrastructure, arguing that traditional ticket management systems are ill-equipped to handle the complexity of modern IT environments. Their multidimensional clustering approach—combining semantic, spatial, and temporal features—achieves a fidelity of 96% while reducing manual workload by 30% [13]. However, they identify a major barrier in the transition from legacy systems to AI-driven frameworks, particularly in preserving institutional knowledge and overcoming organizational resistance.

At the architectural level, Oparin, Bogdanova, and Pashinin (2020) introduce the *HPCSOMAS* framework, which integrates hierarchical control with decentralized peer-to-peer agent networks. This hybrid approach enables high scalability and flexibility in solving complex distributed problems, yet it also highlights the growing complexity of such systems, especially for non-expert users. The lack of high-level abstraction tools remains a key obstacle for broader adoption [20].

Finally, Rosendo et al. (2025) push the boundaries of automation by proposing AI-driven orchestration of scientific workflows through modular agent-based architectures. By combining natural language interfaces with programmable APIs, their system enables adaptive and near real-time coordination of computational resources, significantly accelerating scientific experimentation [23]. Nonetheless, they identify unresolved challenges related to cross-facility integration, as well as limitations imposed by existing security policies that are not designed for interactive or streaming workflows.

Together, these contributions define Cluster 3 as the core of AI-driven optimization in HPC, where scheduling, resource management, and workflow orchestration converge into intelligent, adaptive systems.

Building upon this operational intelligence, the next stage of evolution—Cluster 4—shifts the focus toward closing the loop between system optimization and user support. While Cluster 3 optimizes how resources are allocated and managed, Cluster 4 explores how large language models, retrieval systems, and conversational agents can make these complex systems accessible, interpretable, and increasingly autonomous from the user's perspective.

Building upon the advances in AI-driven scheduling and resource optimization, the final stage in this evolution focuses on closing the loop between system intelligence and user interaction. While Cluster 3 demonstrates how HPC infrastructures can autonomously manage workloads and optimize performance, Cluster 4 shifts attention toward making these capabilities accessible, interpretable, and actionable for users. In this layer, large language models (LLMs), retrieval-augmented generation (RAG), and conversational agents emerge as the key enablers of fully integrated, intelligent support ecosystems—where interaction, automation, and decision-making converge into a unified experience.

At the core of this transition, Rosendo et al. (2025) propose a modular, agent-based architecture that bridges workflow orchestration with natural language interfaces and programmable APIs. Their approach enables adaptive and explainable coordination of scientific workflows in near real time, effectively reducing the reliance on manual intervention [23]. However, they identify persistent barriers in cross-facility integration, particularly due to misaligned security policies and the lack of standards for interactive or streaming workloads—highlighting a structural limitation in current HPC ecosystems.

Complementing this systems-level perspective, Wulf and Meierhofer (2024) explore the role of LLMs in automating technical support tasks. They demonstrate that while models such as GPT-4 are highly effective for low-level cognitive operations (e.g., summarization and translation), more complex reasoning tasks require enhanced architectures like RAG or fine-tuning to mitigate hallucinations. Their work underscores the importance of structured data ecosystems for enabling context-aware AI, yet also reveals a critical gap in large-scale, real-world validation and system integration [25].

Addressing the challenge of fragmented knowledge, Bondapalli et al. (2025) show that HPC documentation—often distributed across heterogeneous formats—can be effectively leveraged through multimodal RAG systems. By preserving the structural integrity of technical content such as code and tables, their approach significantly improves response relevance, achieving a score of 4.7/5.0 [8]. Nevertheless, they identify that retrieval quality remains highly sensitive to noisy or irrelevant context, motivating the need for more advanced techniques such as hybrid search and knowledge graph integration.

From an applied support perspective, Pasupuleti et al. (2025) demonstrate the practical impact of LLM-based chatbots in HPC environments. Their system, enhanced with FAISS and BERT-based semantic retrieval, is capable of autonomously resolving up to 70

Similarly, Shifare et al. (2025) present an integrated solution combining an AI-based Laboratory Manager (ALM) with a RoBERTa-based chatbot, achieving a 98.1

Extending beyond support automation, Yin et al. (2024) introduce the chatHPC architecture, demonstrating how LLMs can be fully integrated into HPC infrastructures to enable programmable and conversational scientific workflows. Their results show significant improvements in model training efficiency and response accuracy through a hybrid combination of supervised fine-tuning and RAG. However, they point out unresolved challenges in scalable model optimization, dataset generation, and hallucination mitigation, as well as the need for tighter integration with simulation frameworks for end-to-end automation [26].

From a communication standpoint, Guo et al. (2021) contribute to the development of more natural interaction modalities by enhancing text-to-speech systems with contextual and semantic encoders. Their approach enables more human-like conversational agents capable of capturing spontaneous speech patterns, which is essential for immersive and effective user support [3]. Complementarily, Ashish et al. (2025) expand the role of AI agents into the domain of human collaboration, demonstrating how virtual mentors powered by lightweight LLMs can facilitate intercultural communication in global HPC projects, although challenges remain in long-term evaluation and curricular integration [14].

Finally, Jain et al. (2024) reinforce the importance of intelligent support infrastructures by showing how multidimensional AI models can significantly improve incident management through more accurate ticket clustering and resolution recommendations. Despite achieving high performance, their work highlights ongoing challenges in adaptability, contextual reasoning, and the transition from legacy systems—issues that directly impact the deployment of LLM-based support ecosystems at scale [9].

Taken together, these contributions define Cluster 4 as the most advanced and integrative layer in the evolution of HPC systems. Here, intelligent agents, LLMs, and retrieval-based architectures converge to create adaptive, context-aware, and increasingly autonomous support ecosystems. Rather than simply assisting users or optimizing resources in isolation, these systems aim to unify interaction, computation, and knowledge into a cohesive framework capable of supporting the full lifecycle of scientific discovery.

New references:

Razma and Jurkus proposed an AI-based framework for the automated classification of IT support requests in enterprise IT Service Management environments. Their approach was designed to streamline ticket triage and routing by leveraging natural language processing and machine learning techniques on a large corpus of historical support tickets [22].

The authors developed a two-stage classification architecture. In the first stage, a Logistic Regression model using TF–IDF representations identified the request type, such as incidents or administrative requests. The predicted request category was then incorporated as an additional feature in the second stage, where multilingual Transformer models, including mBERT and XLM-RoBERTa, were fine-tuned to assign tickets to the appropriate support team [22].

To improve data quality, the dataset underwent extensive preprocessing, including text normalization, HTML tag removal, and duplicate detection through cosine similarity over TF–IDF vectors. Since the ticket repository exhibited a highly imbalanced class distribution, random oversampling was applied during training to enhance the representation of underrepresented categories while preserving realistic class proportions during evaluation [22].

Experimental results showed that Transformer-based models consistently outperformed the traditional Logistic Regression baseline. The most notable improvements were observed in minority classes, where macro F1-score increased from approximately 0.39 to values between 0.49 and 0.51. The study also demonstrated that these gains were achieved while maintaining inference times suitable for operational deployment in enterprise service desks, enabling near real-time ticket processing [22].

Furthermore, interpretability analyses based on LIME and SHAP revealed that Transformer models captured domain-specific semantic relationships more effectively than conventional machine learning methods, which relied primarily on lexical frequency patterns. Based on these findings, the authors concluded that multilingual pre-trained language models combined with data balancing strategies provide a robust and scalable solution for automated ticket classification, contributing to improved operational efficiency and digital transformation initiatives in large organizations [22].

Mani et al. proposed *Agent Assist*, a cognitive support platform designed to automate enterprise IT help desk operations by providing intelligent assistance during issue diagnosis and resolution [15]. The authors recognized that IT support environments generate large volumes of heterogeneous information, including application documentation, historical ticket records, knowledge transfer videos, and user interactions. They argued that support agents frequently receive incomplete or ambiguous problem descriptions, which can lead to incorrect diagnoses, increased resolution times, and higher operational costs [15].

To address these challenges, the authors developed a multi-component architecture that integrates machine learning, information retrieval, and knowledge representation techniques. The system combines question classification models based on Support Vector Machines and Convolutional Neural Networks with a disambiguation module supported by knowledge graphs and similarity-based retrieval using TF–IDF representations. This architecture enables the platform to identify user intent, retrieve relevant information, and generate contextually appropriate recommendations across multiple technical domains [15].

A key contribution of the study is the *Learning on the Job* mechanism, which allows the platform to continuously expand its knowledge base through user feedback and interactions between support agents and end users. Rather than relying exclusively on manually curated knowledge repositories, the system incrementally learns from real operational environments, enabling the acquisition of new expertise with minimal human intervention [15].

The framework also incorporates an image understanding component called *iSight*, which analyzes screenshots containing error messages and visual system information. By extracting relevant details from these artifacts, the module provides additional contextual evidence that assists the system in identifying the root cause of technical issues and improving recommendation quality [15].

The authors evaluated the proposed platform in large-scale enterprise environments and reported successful adoption across more than 650 internal IBM projects. Their findings indicate that the integration of automated question answering, continuous knowledge acquisition, and multimodal information processing can substantially reduce the workload of support personnel while accelerating problem resolution. The study concludes that intelligent support systems should evolve beyond traditional document retrieval approaches toward solutions capable of delivering procedural and context-aware answers, thereby reducing response times and broadening access to specialized technical knowledge throughout organizations [15].

Pagan and Dewi proposed an intelligent virtual assistant for automated IT helpdesk resolution based on Transformer architectures, aiming to improve the efficiency of IT Service Management workflows through automated intent recognition and response generation [21]. Their study focused on the implementation and comparative evaluation of modern language models to determine their suitability for enterprise support environments where rapid and accurate ticket handling is essential [21].

The proposed system was built upon a natural language processing pipeline trained on a dataset of 4,800 annotated IT support requests covering hardware failures, software issues, network connectivity problems, access management requests, and general inquiries. Data preparation involved text normalization, noise reduction, and subword tokenization techniques tailored to each Transformer architecture to ensure effective representation of technical language and user queries [21].

To identify the most appropriate model for automated helpdesk operations, the authors fine-tuned BERT, RoBERTa, and DistilBERT using a supervised learning framework based on the AdamW optimizer and cross-entropy loss. The architecture simultaneously performed intent classification and retrieval-based response generation. Once an intent was identified, the system retrieved relevant resolution notes from a knowledge base through cosine similarity over TF–IDF representations, enabling the delivery of contextually relevant solutions to end users [21].

A confidence-based decision mechanism was incorporated to ensure reliable automation. When the predicted confidence exceeded a predefined threshold, the system automatically resolved the request. Otherwise, the ticket was escalated to a human support agent. This strategy allowed the platform to balance automation and service quality while minimizing the risk of incorrect recommendations [21].

The experimental evaluation demonstrated that RoBERTa achieved the highest predictive performance, reaching a classification accuracy of 93.6% and a weighted F1-score of 0.934. DistilBERT emerged as the most computationally efficient alternative, reducing inference latency by nearly half compared with RoBERTa while maintaining competitive classification results. The authors also reported a ticket deflection rate of 92.8%, indicating that the system was capable of autonomously resolving the majority of routine first-level support requests [21].

Based on these findings, the study concluded that Transformer-based NLP technologies have reached a level of maturity suitable for production deployment in enterprise ITSM environments. The authors emphasized that model selection should be guided by a balance between predictive performance, computational efficiency, and autonomous resolution capability, allowing organizations to align intelligent helpdesk solutions with their operational and infrastructure constraints [21].

Botezatu et al. proposed a data-driven framework for optimizing incident ticket dispatching in enterprise IT support environments by combining ticket clustering and intelligent resource allocation strategies [4]. The study was motivated by the limitations of traditional assignment approaches, which typically rely on predefined skill matrices or manual coordination and often fail to exploit historical performance information when selecting the most suitable technician for a given incident [4].

The authors introduced a multi-view clustering methodology that simultaneously considers the semantic content of incident tickets and historical resolution performance. Rather than grouping tickets solely according to textual similarity, the proposed approach incorporates information about the time required to resolve similar incidents in the past. This combination enables the creation of more meaningful ticket categories in which both problem characteristics and expected service effort are represented. To achieve this objective, the study employed advanced clustering techniques, including Spectral Clustering and an enhanced version of Fuzzy k-means, allowing the identification of specialized incident groups with consistent resolution patterns [4].

Once a new ticket is assigned to a cluster, the system applies a dispatching policy based on empirical performance data. Instead of selecting technicians exclusively according to their declared expertise, the framework identifies the available agent who has historically resolved incidents from the corresponding cluster in the shortest amount of time. This strategy transforms ticket assignment into a data-driven optimization problem that prioritizes operational efficiency and observed performance outcomes [4].

The evaluation was conducted using real-world datasets collected from IBM support centers. Experimental results demonstrated substantial reductions in ticket resolution times, with improvements ranging from 35% to 44% when compared with conventional assignment methods based on manual decision making or static skill-based routing. These findings indicate that historical performance information can significantly enhance the effectiveness of support operations and resource utilization [4].

An important observation reported by the authors is that highly experienced technicians were not always the most efficient choice for every type of incident. In several categories, intermediate-level agents achieved faster resolution times than senior specialists, highlighting the value of identifying expertise through empirical evidence rather than relying solely on organizational hierarchies or certifications. The proposed framework therefore enables organizations to uncover hidden areas of specialization within support teams and allocate resources more effectively [4].

The study concludes that integrating historical performance data into ticket clustering and dispatching decisions offers a practical and scalable mechanism for improving service desk efficiency. Beyond reducing operational costs and resolution times, the approach also provides insights into workforce development by identifying areas where additional training may be required, supporting the continuous improvement of IT support processes [4].

3 Research Questions

Based on the challenges identified in the Kabré HPC support system and the principles of Human-Computer Interaction (HCI), this study investigates both the operational effectiveness of an Intelligent Virtual Agent (IVA) and its impact on user perception, experience, and infrastructure support efficiency. In addition, the study examines the hypothesis that the IVA contributes to improving the overall support workflow by reducing the workload of the infrastructure team through automated handling and intelligent escalation of requests.

The research is guided by the following questions:

1. To what extent can an Intelligent Virtual Agent improve the efficiency of technical support in the Kabré supercomputer environment by reducing perceived waiting times, accelerating information gathering, and facilitating the resolution of common support requests?
2. How effectively can the IVA leverage Natural Language Processing (NLP) and semantic embeddings to understand, classify, and structure user requests in the HPC environment?
3. How accurately can the IVA determine whether a support request can be resolved automatically or should be escalated to the infrastructure support team based on semantic classification and contextual information gathered during interaction?
4. How do users perceive the usability and trustworthiness of the Intelligent Virtual Agent when interacting with it to perform support-related tasks within the HPC environment?
5. How do users perceive the embodied interface of the IVA, including its 3D avatar, speech, and visual expressions, and to what extent do these features contribute to engagement, comfort, frustration reduction, and willingness to seek support through the system?
6. How do user perceptions of the IVA compare with their previous experiences using the traditional ticket-based support system in terms of responsiveness, accessibility, and overall support experience?
7. To what extent does the implementation of the IVA reduce the workload of the Kabré infrastructure support team by handling repetitive requests and improving the quality of escalated cases through structured, context-rich reports?

4 Objectives of the Agent

The Intelligent Virtual Agent (IVA) is designed to act as a first-line support system for the Kabré High-Performance Computing (HPC) infrastructure. Its main objectives are:

New objectives incorporating NLP/embedding:

1. To interact dynamically with users in order to collect the information required to understand the nature, context, and scope of a support request. Through a conversational process, the agent will gather relevant details such as error messages, executed jobs, software configurations, and the resources involved.
2. To leverage Natural Language Processing (NLP) techniques and semantic embeddings to analyze and classify user requests. By capturing the semantic meaning of each interaction, the system will identify patterns associated with common support scenarios and infrastructure-level incidents.
3. To automatically determine whether a request can be resolved by the IVA or should be escalated to the infrastructure support team. This decision will be based on the information collected during the conversation and the semantic classification generated from the embedding representations.
4. To provide real-time assistance for requests that fall within the agent's scope of knowledge, delivering rapid and consistent responses to brief, repetitive, and frequently asked questions related to the use of the HPC environment, job execution procedures, resource access, and other common operational inquiries documented by the infrastructure team.
5. To generate structured and context-rich reports for cases requiring human intervention, ensuring that infrastructure personnel receive all relevant information collected during the interaction, thereby reducing diagnostic effort and resolution times.
6. To reduce the workload of the human support team by automating the handling of repetitive and frequently asked questions while reserving human expertise for complex, specialized, or critical incidents.
7. To improve the overall user experience within the HPC environment by providing clear, adaptive, and accessible guidance tailored to different user profiles, including researchers, students, and system administrators.

As a result, the implementation of the IVA benefits both Kabré users and the infrastructure support team. Users can obtain immediate assistance for common questions, reducing waiting times and making it easier to use the HPC platform effectively. At the same time, the infrastructure team can focus on more complex issues, since repetitive inquiries are handled automatically and escalated cases include the necessary context gathered during the interaction. Overall, the IVA helps improve the support process, enhances the user experience, and contributes to a more efficient operation of the Kabré platform.

5 Resource Management and Coexistence in HPC Computing

The deployment of the Intelligent Virtual Agent (IVA) within the Kabré High-Performance Computing (HPC) infrastructure must be carefully managed to ensure coexistence with ongoing scientific workloads. Since Kabré is a shared computational resource used by researchers from multiple scientific domains, the execution of AI inference services should not interfere with the allocation of resources required for production jobs.

The IVA backend and its associated language models will be deployed on dedicated computational resources within the Kabré environment. Depending on resource availability and operational requirements, inference services may be executed on dedicated CPU or GPU nodes reserved for support and service-oriented workloads. This approach prevents competition with scientific computing jobs submitted by researchers.

Resource allocation and execution management will be coordinated with the CeNAT infrastructure team, particularly with the administrators responsible for cluster operations. Special consideration will be given to scheduling policies and resource quotas to ensure compliance with existing operational procedures.

Kabré uses the Simple Linux Utility for Resource Management (Slurm) as its workload manager and job scheduler. Slurm is an open-source resource management system widely adopted in HPC environments for allocating computational resources, scheduling jobs, monitoring execution, and enforcing usage policies. Through Slurm, users submit computational jobs to queues, which are then executed according to resource availability and scheduling priorities.

The deployment and testing of the IVA will therefore be coordinated with the cluster administration team to ensure that any processes requiring computational resources are properly scheduled and monitored through Slurm. This coordination will help guarantee system stability, prevent resource contention, and maintain the quality of service provided to the scientific community using Kabré.

Furthermore, any future deployment involving larger language models or GPU-accelerated inference services will require additional evaluation of resource consumption, scheduling impact, and long-term operational sustainability within the shared HPC environment.

6 Preliminary Technical Architecture

The architecture of the Intelligent Virtual Agent (IVA) is designed as an integrated system combining large language models, voice services, and an interactive interface to support the technical assistance process within the Kabré HPC environment. The design is motivated by the heterogeneity of the user community, which implies that interactions cannot follow a single uniform behavior. Instead, the agent is expected to adapt its responses according to predefined user clusters such as social sciences, natural sciences, or computational disciplines, an approach consistent with recent work on LLM-based support systems in HPC environments [25] and findings in conversational agent research highlighting the importance of user characteristics and adaptive interaction design in socio-technical systems [7]. This adaptation affects both the level of technical detail and the explanatory style used in responses.

At the core of the system, the Gemma 4 E4B model is used, deployed within the Kabré infrastructure. This model is selected due to its ability to operate efficiently in high-performance computing environments and its strong performance in natural language understanding and generation tasks. These characteristics make it suitable for handling a wide range of technical support queries, from basic system usage to more complex issues such as job execution, software installation, and resource management.

Interaction with the system is based exclusively on textual input from the user. Users communicate with the agent only through text, without any speech input capability. However, the system incorporates VoxCPM2 as a text-to-speech (TTS) engine for the agent's output. In this configuration, the language model generates textual responses that are subsequently converted into speech, enabling a multimodal output experience while maintaining a simple and consistent text-only input interface. The use of speech synthesis in this context is motivated by advances in conversational neural TTS systems, which show that incorporating conversational context and semantic awareness significantly improves prosody naturalness and perceived realism in voice agents [12]. Such approaches demonstrate that high-quality conversational TTS benefits from modeling dialogue context and utterance-level dependencies, leading to more natural and contextually appropriate speech generation.

The user interface is developed using Unity, enabling a real-time interactive environment for communication with the agent. This interface serves as the main point of interaction between users and the support system, complementing the traditional ticket-based workflow. The use of an avatar-based representation of the agent is intended to make the interaction more intuitive and accessible, particularly for less technical users. This design choice is consistent with research on interactive agent-based avatar systems, where embodied virtual agents have been shown to improve usability and engagement by combining visual representation with structured behavioral models [5].

Additionally, the system may incorporate facial animation technologies such as the NVIDIA LipSync model to improve synchronization between generated speech and visual representation. This component is considered optional and is intended to enhance the naturalness of the interaction experience.

The system is designed to run primarily within the Kabré infrastructure, leveraging its computational resources for both the language model and backend services. The IVA will be accessible exclusively through the official Kabré website, operating as an online service integrated into the institutional portal, where users can interact with the agent directly from their web browser.

Overall, this architecture is intended to address the main limitations identified in the current support system, including ticket overload, user heterogeneity, and long response times. Rather than replacing human support, the system is designed to complement it by automating repetitive queries while escalating complex cases to human experts when necessary.

7 Methodology

7.1 Type of Study:

This research is structured as a controlled empirical user study employing a within-subjects design. The study evaluates the Intelligent Virtual Agent (IVA) based on the same users who have previously used the conventional ticketing system of the Kabré supercomputer, leveraging their direct experience with the traditional support service as a reference for assessing potential improvements in efficiency, usability, and interaction quality.

The evaluation of variables such as perceived usability, system trust, and reduction of frustration is grounded in genuine cognitive and emotional responses provided by real end-users when facing authentic technical scenarios. Since participants are already familiar with the ticketing system, their judgments enable a contextualized comparison based on prior experience, which strengthens the ecological validity of the study and allows for a more precise identification of perceived changes in system interaction.

Additionally, the technical performance of the system will be complemented by an evaluation conducted by the CeNAT infrastructure team, who will assess the IVA's ability to correctly classify user queries and appropriately route requests that cannot be resolved by the agent to human support. In particular, this evaluation will focus on the accuracy of the natural language processing (NLP) component in intent detection, the correct escalation of "non-automatable tickets" to human channels, and the quality of the responses generated by the agent in real-world technical support scenarios.

As future work, the study will be extended by introducing two additional experimental conditions to isolate the impact of interface components: a non-avatar agent (purely text-based interface) and an agent with a 3D animated avatar. This comparison will enable a more precise evaluation of the specific contribution of visual representation and animation to variables such as trust, cognitive load, and user experience.

7.2 Participants:

- *Inclusion Criteria:*
  Participants must be currently registered and active users of the Kabré High-Performance Computing (HPC) infrastructure. To ensure a comprehensive evaluation of the Intelligent Virtual Agent (IVA), the sample will incorporate users with diverse levels of technical expertise, ranging from domain scientists

(e.g., biologists, chemists, and physicists) who rely on the cluster for scientific computing tasks, to advanced computer scientists and system engineers.

Additionally, participants must have previously submitted and managed at least one technical support request through the traditional CeNAT ticketing system. This prior experience is a mandatory requirement, as it provides the contextual knowledge necessary to assess the IVA in relation to the existing support workflow.

From a statistical sampling perspective, the target population consists of approximately 300 active users of the Kabré HPC infrastructure. Using standard sample size estimation for finite populations under conservative assumptions (95% confidence level, $p = 0.5$, and a 5% margin of error), the recommended sample size would be approximately 169 participants. However, this value corresponds to large-scale survey studies and does not account for the characteristics of controlled experimental evaluations.

Due to the exploratory nature of this research, the study will initially be conducted as a pilot evaluation. At the time of writing, approximately 10 active Kabré users have verbally indicated their interest in participating in the study and are expected to form the initial evaluation group. Since all of them have prior experience with both the HPC infrastructure and the traditional CeNAT ticketing system, they are well positioned to provide meaningful feedback regarding the performance, usability, and overall effectiveness of the Intelligent Virtual Agent.

– *Exclusion Criteria:*

  Individuals with no prior experience utilizing the Kabré infrastructure or those who have never interacted with the traditional technical support ticketing system will be excluded, as they lack the necessary context to evaluate the perceived efficiency, usability, trust, and reduction of frustration associated with the Intelligent Virtual Agent.

7.3 Metrics

To evaluate the effectiveness of the Intelligent Virtual Agent (IVA) in the Kabré supercomputer support environment, both **objective performance metrics** and **subjective user-centered metrics** are collected.

Objective Metrics

Task Resolution Time (seconds). Measures the elapsed time from the moment a participant submits the first query until the assigned task is successfully completed. This metric is directly associated with **RQ1**, as it provides an empirical indicator of support efficiency and enables comparison with the perceived waiting times reported by participants based on their previous experience with the traditional ticketing system.

Task Success Rate (%). Represents whether participants successfully complete the assigned support task without external assistance. Success is recorded as a binary outcome, indicating either task completion or task non-completion, and is subsequently aggregated across participants. This metric supports **RQ1** by evaluating the practical effectiveness of the Intelligent Virtual Agent in resolving technical issues commonly encountered within the Kabré infrastructure.

Number of Interaction Turns. Counts the total number of exchanges between the participant and the IVA during task completion. Lower numbers may indicate more efficient guidance and clearer support instructions. This metric complements **RQ1** by providing an additional measure of interaction efficiency.

Subjective Metrics

Agent Response Quality Questionnaire (ARQx). The ARQx instrument evaluates perceived efficiency, competence, clarity, and usefulness of the responses generated by the Intelligent Virtual Agent. The resulting scores provide a quantitative measure of support quality as perceived by the participants. This metric is directly related to **RQ1**, which investigates whether the Intelligent Virtual Agent improves technical support efficiency relative to the traditional ticket-based support system.

System Usability Scale (SUS). The System Usability Scale (SUS) provides a standardized measure of perceived usability through a ten-item questionnaire. Responses are transformed into a score ranging from 0 to 100, where higher values indicate greater usability and ease of interaction. This metric is associated with **RQ2** because it evaluates how effectively users can interact with the Intelligent Virtual Agent while performing technical tasks related to software configuration and system management.

Perceived Trust in the IVA. Trust is measured through questionnaire items included in the post-interaction evaluation, as well as through qualitative feedback collected after task completion. These measures assess the level of confidence that participants place in the recommendations provided by the system and in its autonomous operation. This metric is directly associated with **RQ2**, which examines the influence of autonomous agent behavior on user trust.

NASA Task Load Index (NASA-TLX). Two dimensions of the NASA Task Load Index (NASA-TLX) are employed in this study: Mental Demand and Frustration Level. The Mental Demand dimension measures the cognitive effort required to understand and complete the assigned task, whereas the Frustration Level dimension evaluates feelings of stress, irritation, insecurity, or discouragement experienced during the interaction. These measures are directly related to **RQ3**, as they provide evidence regarding whether the Intelligent Virtual Agent reduces cognitive load and frustration when users interpret technical support information and troubleshoot complex system issues.

Godspeed Questionnaire Series. The Godspeed Questionnaire is used to assess user perceptions of the animated agent, with particular emphasis on anthropomorphism, likeability, perceived intelligence, and animacy. The scores obtained from these dimensions provide evidence regarding the influence of the visual appearance and vocal characteristics of the avatar on the overall user experience. This metric is directly associated with **RQ3**, which investigates whether the presence of an animated and expressive virtual agent contributes to reducing frustration and improving the interaction experience.

7.4 Evaluation Protocol Design and a Consistency Matrix

1. Definition of Conditions.

   Experimental Condition:
   The experimental condition introduces participants to the Intelligent Virtual Agent integrated into the Kabré supercomputer web portal. Participants interact with the agent through a browser-based interface featuring a 3D animated avatar developed in Unity, which provides visual expressions and synchronized speech to support communication.
   The system uses an intelligent routing mechanism that operates entirely in the background: when a user submits a query, the backend automatically identifies its nature and context, such as routine software installation requests or complex job management issues, and routes it to the most appropriate processing component.
   The IVA is designed to assist users in resolving common technical support scenarios within the Kabré infrastructure. Users interact with the system by submitting natural language queries in plain text, and the agent responds using a combination of structured textual explanations and voice synthesis powered by the *VoxCPM2* engine.
   During the experiment, participants complete a set of pre-defined and standardized technical tasks representative of real support cases in the Kabré environment, including job submission issues (Slurm), software environment configuration using modules, and interpretation of system error logs. All participants perform the tasks concurrently under the same experimental session to ensure consistent exposure conditions.
   The system operates in real time and is evaluated under controlled laboratory conditions, ensuring that all participants are exposed to the same set of tasks and interaction flow. No human support staff intervenes during the interaction, allowing the assessment of the IVA as a fully automated support interface.

   Baseline Definition (Historical Ticketing Experience):
   The baseline condition in this study is not a concurrently administered experimental group, but rather a retrospective and experiential baseline derived from participants' prior interactions with the Kabré technical support

ticketing system. All participants are active users of the supercomputing infrastructure who have previously submitted and managed support tickets through the CeNAT helpdesk platform. As a result, they possess an internalized reference of system performance, including perceived waiting times, response delays, and interaction friction associated with the traditional asynchronous support workflow.

This study therefore adopts a within-subject design in which participants are exposed only to the Intelligent Virtual Agent (IVA), and all comparative judgments regarding efficiency, usability, and frustration are anchored against their prior real-world experience with the ticketing system.

Justification:

The selection of this baseline is fundamental to isolate and quantify the empirical impact of the Intelligent Virtual Agent's architecture and features on system performance and user experience. This study does not use a concurrent control group, but instead adopts a within-subject design based on a retrospective baseline derived from participants' prior experience with the Kabré technical support ticketing system.

This approach enables comparison between the interaction with the Intelligent Virtual Agent and the users' previously internalized experience, including perceived waiting times, response delays, and interaction friction accumulated through real-world use of the traditional support system.

Methodologically, the use of standardized technical tasks ensures controlled interaction with the automated system, such that response time, success rate, and interaction efficiency metrics reflect solely the performance of the agent. This eliminates the influence of delays inherent to the human-operated asynchronous support system and allows a more direct evaluation of the automated interface.

Furthermore, this design enables the assessment of whether the visual and auditory components of the interface contribute to reduced frustration and improved perceived usability when contrasted with users' prior experience with the traditional ticketing system. Consequently, the study provides evidence regarding the potential of the Intelligent Virtual Agent to improve efficiency and user experience in high-performance computing support environments.

2. Methodological Consistency Matrix.

   The experimental design and its operational alignment are structured in Table 2

3. Adaptation of the Researcher's Protocol.

   The experimental protocol was developed and implemented in HTML, integrating study identification, informed consent, randomized group assignment, task instructions, post-interaction questionnaires, and data anonymiza-

Research Questions	**Variable or Construct**	**Validated Instrument**	**Associated Task**
RQ1: *To what extent can an Intelligent Virtual Agent improve the efficiency of technical support in the Kabré supercomputer environment, specifically minimizing user-side waiting latencies and optimizing request processing compared to traditional ticketing?*	Perceived Efficiency, Agent Competence, and Support Resolution Time	ARQx and objective metrics of resolution time	The participant interacts with the IVA to resolve a Slurm-related support scenario reflecting common historical ticketing cases in the Kabré environment, such as correcting an incorrect job submission parameter or checking the execution status of a job in the queue.
RQ2: *What is the perceived usability of an automated IVA across a user base with diverse levels of technical expertise, and how does the agent's operational autonomy influence user trust when navigating complex supercomputing tasks?*	System Usability, Trust, and Expertise-Based Experience	System Usability Scale (SUS) and qualitative user feedback	The participant uses the IVA to locate, load, and configure a scientific software dependency using environment modules, following the agent's guidance with minimal external assistance in order to evaluate perceived usability, trust, and system autonomy.
RQ3: *Does interacting with a 3D animated avatar that speaks and shows expressions actually help reduce user frustration, or does it feel like an unnecessary distraction compared to a simple, text-only chat?*	Perceived Frustration, Mental Demand, and Interface Animacy	NASA-TLX (Frustration and Mental Demand subscales) and Godspeed Questionnaire	The participant is presented with a complex and ambiguous system error log related to disk space or file access restrictions. The agent assists in interpreting the issue, enabling assessment of whether animated and vocal feedback reduces perceived frustration and cognitive load compared to raw textual logs.

tion policies. All materials can be accessed in the project repository.

4. Theoretical Justification in Human-Computer Interaction (HCI).

   The architectural and interaction design choices implemented in the Intelligent Virtual Agent for the Kabré supercomputer are grounded in foundational frameworks of HCI and social psychology, aligned with the methodological framework of the study.
   The design of this virtual agent was not conceived as a mere visual enhancement, but as a solution to real interaction problems within the computing cluster. According to [6], the use of a stylized humanoid avatar instead of a

plain text-based chat interface fulfills a key functional role. The moderate facial expressions and lip-syncing achieved through the VoxCPM2 engine help guide the conversation in a more natural way. By also communicating through non-verbal cues, the interface regulates dialogue pacing and makes support instructions easier to process, reducing the user's cognitive load and avoiding discomfort associated with the *uncanny valley* thanks to a balanced visual design.

This visual support is directly complemented by the management of the user's emotional state. Based on the strategies described by [1], incorporating empathetic responses and a more natural conversational style is essential to reducing frustration in a highly complex environment such as supercomputing. The main advantage here is immediacy: unlike traditional ticketing systems, where a researcher may wait hours or days for a technical response in full uncertainty, the agent acts as a real-time companion precisely when the issue occurs. When code fails to compile or storage becomes blocked, the agent responds instantly, acknowledges the user's stress, provides an immediate solution, builds trust, and prevents task abandonment.

Finally, this trust does not depend only on what the agent says or when it says it, but also on the authority projected through its visual presence. Although the Proteus Effect analyzed by [**yee**] explains how a person's behavior changes depending on their own avatar representation, its underlying psychological principles demonstrate that the visual design of a character strongly shapes the expectations of the observer. By unifying the agent's identity with the cultural symbols of CeNAT's computing nodes (Nukwa, Kurá, and Dribe), the interface naturally conveys a sense of institutional competence and technical expertise. This directly influences the researcher's confidence, increasing trust in the system and improving perceived usability when configuring complex scientific variables and dependencies.

7.5 Exploratory Data Analysis

The data extracted from the Zammad SQL database of the Kabré supercomputer ticketing system are analyzed in order to identify frequent user queries and classify request types. The objective of this analysis is to derive meaningful support categories that enable the assignment of specialized agents to each type of request.

7.6 Current state of the intelligent virtual agent

- **Description of the prototype implementation status. Rigging and Animation of the Avatar in Blender.**
  For the visual representation of the agent, the three-dimensional model *CHAMAN TX*, developed by Isaac Villarreal, was selected as a free asset from the Sketchfab platform under a Creative Commons Attribution (CC-BY) license. In its original state, this asset consisted of a static mesh lacking

facial rigging, an internal skeletal structure, or deformation morphs, factors that significantly limited its expressiveness and naturalism. With the objective of providing the agent with communicative functionality, an exhaustive technical intervention was carried out within the *Blender* environment, which was fundamental for transforming the model's base geometry.

In the first instance, utilizing *Blender*'s rigging tools, an internal skeletal system was implemented, designed to provide the model with controlled mobility in key regions such as the chest, upper extremities, and head. This skeletal structure allowed for the establishment of an animation cycle focused on simulating breathing, thereby granting an organic dynamism to the character.

Complementary to the skeletal structure, a topological editing process was executed to isolate the labial structure of the original model. On this geometry, *Shape Keys* were parameterized within *Blender*—deformation morphs specifically configured to define the opening of the oral cavity. This procedure enabled the definitive transition from a static three-dimensional asset to a fully articulated model, capable of exhibiting both cyclic body movements and dynamic gestural responses.

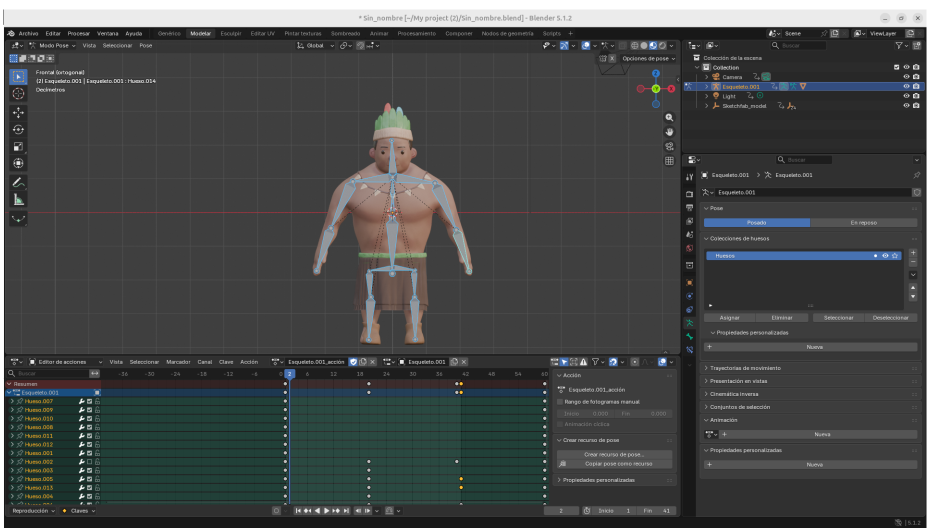

Fig. 1. Implemented skeleton and breathing animation of the avatar in *Blender*.

Integration of the Avatar into Unity: Interface, Animation Control, and Backend Communication.

Following the preparation of the three-dimensional model, the resulting FBX file was imported into the Unity development engine to construct the main

Fig. 2. Preparation of the mouth geometry in *Blender*, involving topological isolation and Shape Key parameterization for future lip-sync functionality.

scene. During this phase, the material properties of the imported avatar were customized to align with the institutional color palette of Kabré. This was achieved by modifying the *Albedo* properties of the avatar's *Materials*, specifically applying the following RGB configurations: blue (23, 23, 155), red (242, 10, 10), orange (255, 69, 0), and yellow (218, 165, 32), alongside white accents, to establish a cohesive visual identity for the agent.

To manage user interaction, a comprehensive two-dimensional Graphical User Interface (GUI) was engineered utilizing Unity's *Canvas* component, which serves as the root rendering space for all UI elements. The interface architecture heavily relies on the *TextMeshPro* (*TMP*) library to ensure optimized rendering and advanced typography control. The UI layout comprises a *TMP_InputField* designated for capturing user text queries, an interactive *Button* component configured with an *onClick()* event listener mapped to trigger the data submission function, and a dedicated *TextMeshProUGUI* element (identified as *TextoRespuesta*) responsible for displaying the transcribed responses from the agent. The entire client-side operational logic, along with the event handling processes, was centralized within a hierarchical *GameObject* named "GestorIA", which serves as the primary container for the custom control scripts and the *AudioSource* component.

Communication between the graphical environment and the artificial intelligence engine is administered by the *ChatClient.cs* script. This module is responsible for capturing the text strings entered by the user through the input field, packaging them into a JSON data structure, and transmitting them asynchronously via HTTP POST requests to a local server. Upon resolution of the request, the system processes the response payload, immediately updates the visual text component in the interface, and utilizes the *UnityWebRequestMultimedia* class to download and prepare the audio asset corresponding to the agent's voice.

Concurrently, the avatar's facial animation is governed by the *AvatarController.cs* script. This controller ensures dynamic lip-synchronization (*lip-sync*) through real-time acoustic analysis. During the engine's *LateUpdate* phase, the script extracts a block of digital samples from the playing *AudioSource* and calculates the amplitude envelope using the Root Mean Square (RMS) technique. This acoustic value is subsequently multiplied by a sensitivity variable, and the result is directly mapped as the weight of the labial deformation morph ("Clave 1"). This mathematical approach allows the opening of the 3D model's oral cavity to respond with precise accuracy to the intensity variations of the emitted audio.

The computational core of the system resides in a backend architecture implemented in Python using the FastAPI framework, which exposes asynchronous communication routes and statically serves the generated audio files. The cognitive nucleus of the project is founded on the integration of the Google Gemini API, specifically implementing the *gemini-2.5-flash-lite* large language model. Connection to this service is established through authentication with a private access key, allowing the server to transmit user queries and receive the textual content generated by the artificial intelligence efficiently and with low latency.

Once the Gemini model generates the text string corresponding to the response, the server delegates the phonetic synthesis to the *edge-tts* Python library. To ensure a consistent and natural vocal identity, the acoustic synthesis was configured using the specific neural voice model "es-MX-JorgeNeural". This tool processes the textual response and generates an audio file in MP3 format. Finally, the server temporarily stores this acoustic asset in a local directory and responds to the original request from Unity by sending a JSON data package, which contains both the textual transcription of the response and the static URL of the sound file, thus completing the full cycle of the conversational agent's multimodal interaction.

List of Implemented Functionalities:

- The three-dimensional rendering of the avatar is adapted to the institutional visual identity and equipped with an internal *rigging* system that executes a cyclical breathing animation during periods of inactivity to provide an organic representation.
- Dynamic lip-synchronization (*lip-sync*) is performed through real-time acoustic analysis by calculating the root mean square (RMS) of the playing audio to govern the deformation morphs (*Shape Keys*), guaranteeing precise oral articulation synchronized with the voice.
- A multimodal cognitive engine integrates a large language model for the baseline processing of queries, which is coupled with a neural *Text-to-Speech* engine to emit verbal responses.

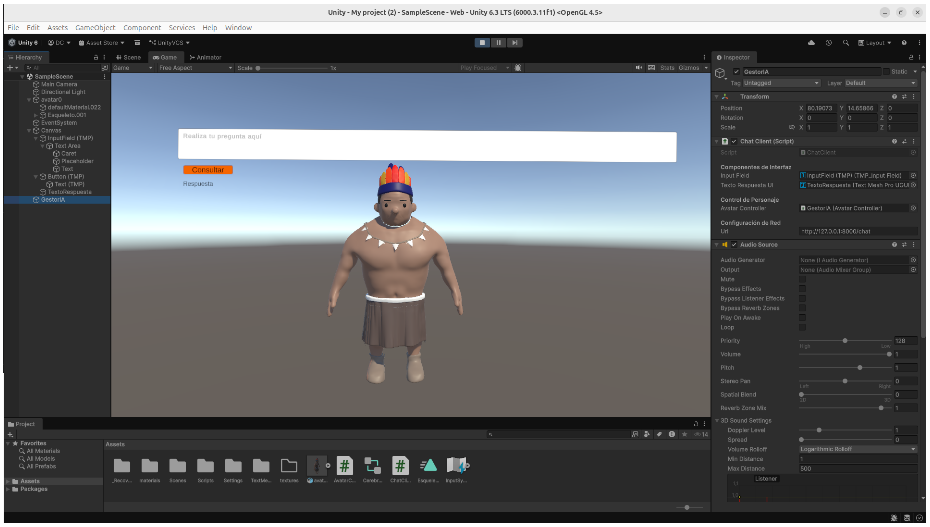

Fig. 3. Transfer of the avatar from Blender to Unity through FBX export, adaptation of the avatar colors to the CeNAT visual identity, and implementation of functionalities including lip-sync synchronization, user text input controls, and display of the language model responses.

List of Pending Functionalities:

- The deployment of a web environment requires the exportation of the graphical client and the configuration of the network architecture to allow interactive access to the agent directly from web browsers.
- The integration of the Natural Language Processing (NLP) engine involves the implementation of semantic *embeddings* in the avatar's processing flow to capture the context of each interaction, allowing incident classification aligned with patterns found in the clustering of the historical support ticket database.
- The active collection of technical parameters will be achieved through the development of a directed conversational flow, enabling the agent to dynamically interrogate the user to extract critical operational variables such as specific error messages, software configurations, or details of the executed job.
- Automated incident triage will utilize a semantic decision algorithm to determine whether a query is a repetitive doubt resolvable in real-time by the agent itself, or if its technical complexity demands escalation.
- The generation of structured reports for escalation will package the technical information gathered by the agent to build context-rich summaries, thereby minimizing the diagnostic effort required by the technical staff upon receiving an escalated case.

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